

$$\begin{bmatrix} \tilde{\mathbf{A}} \end{bmatrix} \begin{bmatrix} \tilde{\mathbf{e}} \end{bmatrix} = \xi_1 \begin{bmatrix} \tilde{\mathbf{b}}_1 \end{bmatrix} + \dots + \xi_P \begin{bmatrix} \tilde{\mathbf{b}}_P \end{bmatrix}$$

$$\check{\mathbf{E}}_{\{\hat{\theta}, \hat{\phi}\}} = \eta_{Q_\theta Q_\phi} \begin{bmatrix} \tilde{\mathbf{c}}^H_{Q_\theta Q_\phi \{\hat{\theta}, \hat{\phi}\}} \end{bmatrix} + \dots + \eta_{Q_\theta Q_\phi} \begin{bmatrix} \tilde{\mathbf{c}}^H_{Q_\theta Q_\phi \{\hat{\theta}, \hat{\phi}\}} \end{bmatrix} \begin{bmatrix} \tilde{\mathbf{e}} \end{bmatrix}$$